

Sanyam Katoch

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Skills

Languages:	Python, Java, C, C++, HTML, CSS
Tools/Platforms:	GitHub, Jupyter Notebook, Streamlit, Google Cloud, Kali, MySQL
Frameworks/Libraries:	Keras, Scikit-learn, Pandas, NumPy, Seaborn, Matplotlib, OpenCV
Soft Skills:	Strategic Thinker, Collaborative, Deadline Driven, Fast Learner

Projects

BotShiled — CAPTCHA Refinement System

Nov'25

- Built a machine learning-driven CAPTCHA system using Python that generates human-friendly yet bot-resistant CAPTCHAs.
- Developed a CNN with TensorFlow/Keras to classify CAPTCHA difficulty and integrated it with a dynamic generator in Streamlit to optimize noise and distortion.
- Delivered an end-to-end pipeline with 6000+ labeled CAPTCHA images and an interactive demo for real-time difficulty adjustment.

SegVision — Image Segmentation System

May'24

- Improved precision and overall performance of image segmentation using PSO (Particle Swarm Optimization).
- Developed an AI-driven segmentation model that uses PSO to enhance feature extraction and object detection accuracy. Utilized Python, NumPy, and OpenCV to optimize segmentation boundaries through iterative particle updates.
- Achieved 70% accuracy in correctly differentiating complex image regions.

Training

PEP Program (DSA Module) — byteXL

Jun'25 – Jul'25

- Learned and practiced Data Structures and Algorithms, enhancing problem-solving skills and coding efficiency. Gained a strong understanding of the core algorithmic techniques through structured hands-on sessions.

Certificates

Data Structures and Algorithms by Dr. Abdul Bari Udemy	Dec'24
The Bits and Bytes of Computer Networking Google	Sep'24
Software Development Processes and Methodologies University of Minnesota	Apr'24
Deep Learning Fundamentals with Keras edX	Mar'24
Master AI and AI Generative Tools Udemy	Dec'23

Extra-Curricular Activities

Social work Volunteer — Supported children with mental disabilities in an NGO by helping them with academic learning and developmental activities.

Core member of the Technical Club (Palo Alto Networks) — Engaged in coding/tech club events, participated in workshops, hackathons, and peer learning sessions.

Hacker-a-thon — Competed in a Web + AI hackathon, contributing to innovative solutions and collaborative problem-solving.

Football

Cricket

Education

Lovely Professional University

Phagwara, Punjab

Bachelor of Technology — Computer Science and Engineering (AI & ML)

Aug'23 – Present

CGPA — 8.0

Mount Carmel Sr. Sec. School

Palampur, Himachal Pradesh

Intermediate — 70%

Apr'21 – Mar'23

Mount Carmel Sr. Sec. School

Palampur, Himachal Pradesh

Matriculation — 94%

Apr'19 – Mar'21